

EXTRACTION LAB · BIOMASS FLOW

High-Level 7-Phase Process Overview
Charge Through THCA / HTE Pour · 34 Steps · 4 Decision Gates

P1

CHARGE, REACTOR PREP
& PRIMARY EXTRACTION
Steps 1 - 14

- Scan / enter biomass lot ID
- Confirm lot readiness
- Record charge & select reactor
- Load biomass / baskets
- Confirm reactor & tank readiness
- Confirm vacuum down 🚩 Critical
- Record primary solvent charge 🚩
- Primary soak (30 min) + mixer (5 min)
- Confirm basket filter clear 🚩
- Nitrogen pressurization → Begin recovery

→ Phase 2

P2

FLUSH CYCLE,
RECOVERY & FINAL PURGE
Steps 15 - 25

- Begin flush cycle
- Verify flush temps ≤ −40 °F 🚩 Critical
- Post temperature evidence
- Record flush solvent charge 🚩
- Flush soak (10 min) + agitation
- Decision 3 — Flow Resumed?
 - YES → Start final purge
 - NO → Adjust flow ⚡ re-check
- Start / stop final purge
- Decision 4 — Final Clarity Confirmed?
 - YES → Shutdown checklist
 - NO → Resume purge ⚡ re-check

→ Phase 3

P3

BOOTH SHUTDOWN
& RUN COMPLETION
Steps 26 - 28

- Complete shutdown checklist 🚩 Critical
 - ① Recovery inlets closed
 - ② Filtration pump-down started
 - ③ Nitrogen off
 - ④ Dewax inlet closed
- Mark run complete
- Select post-extraction pathway
 - 100 lb pot pour OR 200 lb minor run

→ Phase 4

P4

WET THCA / HTE POUR
& HANDOFF
Steps 29 - 34

- Start post-extraction handoff
- Prepare receiving containers & labels
- Pour / collect wet HTE 🚩 Critical
- Pour / collect wet THCA 🚩 Critical
- Confirm initial outputs
 - Both wet gram values required 🚩
- Clean / reset extraction equipment

→ Phase 5

P5

DECISION TREE
OVERVIEW
4 Decision Gates

- D1 — Lot Ready To Charge?
 - PASS → Record charge & load reactor
 - FAIL → Resolve readiness issue ⚡
- D2 — Flush Temps Acceptable?
 - PASS → Charge flush solvent
 - FAIL → Correct temp / evidence ⚡
- D3 — Flow Resumed?
 - YES → Final purge
 - NO → Adjust flow ⚡
- D4 — Final Clarity Confirmed?
 - YES → Shutdown checklist
 - NO → Resume purge ⚡

→ Phase 6

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ROLES, TIMING &
KEY CONTROLS
Reference

- EXT — Extractor (all phases)
 - SUP — Supervisor (exceptions / out-of-range)
 - DOWN — Downstream operator (Phase 4)
 - MAINT — Maintenance (Phase 1 support)
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- Primary soak: 30 min target
 - Mixer / agitation: 5 min target
 - Flush soak: 10 min target
 - Final purge: optional / settings-driven

→ Phase 7

P7

PROCESS GAPS &
SUGGESTED IMPROVEMENTS
SOP Alignment Review

- Add laminated "Run Card" matching app steps
- Group tablet into 4 big phases on-screen
- Visual progress rail showing branch loops
- Make prep & cleaning visible in app
- Display role initials on each step
- Confirm cleaning / readiness ownership
- Confirm −40 °F hard-stop vs. warning
- Confirm Slack vs. app photo as evidence
- Confirm output container requirements
- Confirm "run complete" definition